Global Budget Transparency

A Data Mining Approach to Identifying Socio-Economic Correlates of Fiscal Accountability

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I. Abstract

This study examines glo bal budget transparency using data from the Open Budget Survey available on the World Bank Data360 platform. It focuses on identifying patterns and relationships between transparency, public participation, and budget oversight.

The dataset includes over 100 countries across multiple years. Data preprocessing was performed to address missing values, remove inconsistencies, and standardize formats. Using Python, Pandas, and Jupyter Notebook, the study applied exploratory data analysis and correlation analysis.

Results show that transparency varies widely across countries and regions. A moderate positive relationship was observed between transparency and oversight, indicating that countries with more transparent budgets tend to have stronger monitoring systems. The analysis also shows a decline in transparency between 2021 and 2023, based on the latest available data. These findings suggest that fiscal transparency is influenced by institutional strength and political conditions.

II. Introduction

Budget transparency refers to how openly governments share information about how public funds are collected and spent. It is an important part of good governance because it promotes accountability, builds public trust, and helps ensure that resources are used effectively.

The Open Budget Survey, developed by the International Budget Partnership, provides standardized measures of transparency, public participation, and oversight. These indicators allow comparisons across countries and over time.

This study aims to:

- Examine global trends in budget transparency
- Compare transparency across regions
- Analyze relationships between transparency, participation, and oversight
- Investigate changes at the country level

III. Methodology

a. Data Preparation

The dataset was obtained from World Bank Data360 and cleaned to ensure accuracy and consistency. The following steps were performed:

- Removal of irrelevant or incomplete variables
- Handling of missing values
- Standardization of column names and formats
- Exclusion of non-comparable entries (e.g., pilot data)

These steps ensured that the dataset was consistent and suitable for analysis.

b. Tools Used
- **Python** for data processing
- **Pandas** for data manipulation
- **Matplotlib** for visualization
- **Jupyter Notebook** for workflow execution

c. Analysis Methods
- Exploratory Data Analysis (EDA): Used to examine trends, distributions, and detect outliers through summary statistics and visualizations
- Correlation Analysis: Pearson correlation was used to measure relationships between transparency, participation, and oversight

IV. Results and Discussion

a. Global Trend (2006–2025)

Global transparency shows a gradual improvement from 2006 to 2021, indicating long-term progress in fiscal accountability. However, a noticeable decline is observed in 2023 based on the most recent available data.

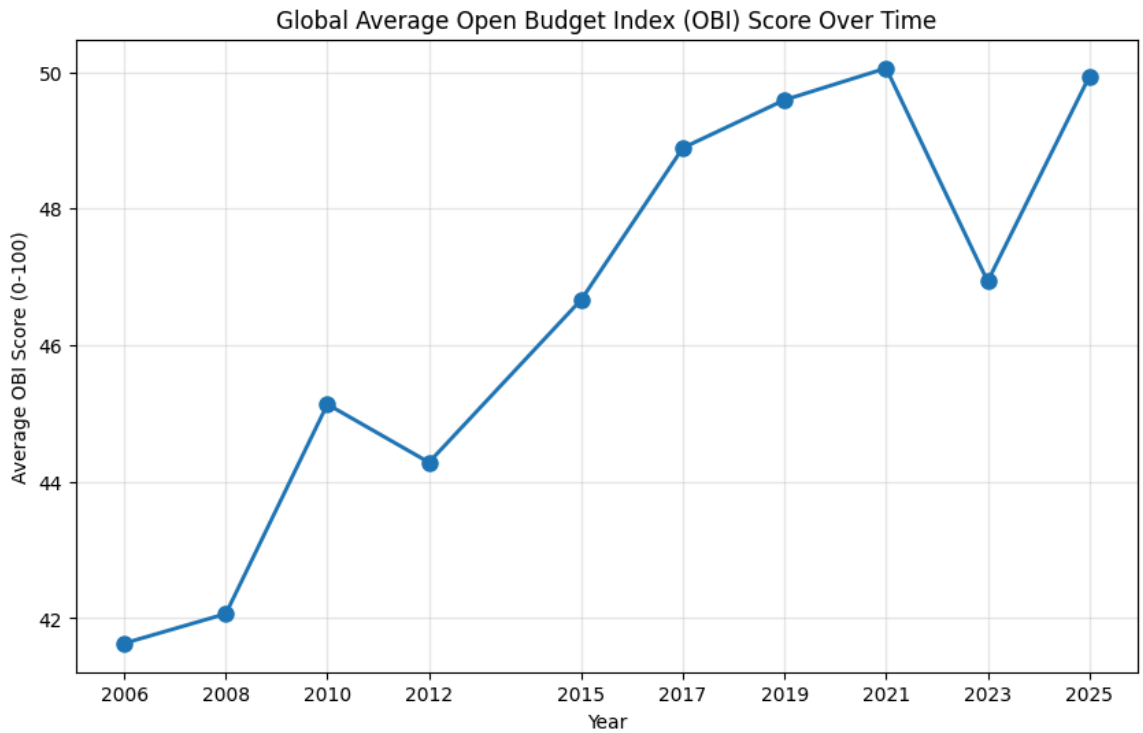

Figure 1. Global Average OBI Over Time

Key insights:

- Steady improvement from 2006 to 2021
- Significant drop in 2023
- Partial recovery in 2025, though still below peak levels

b. Regional Comparison

These differences suggest that governance systems, institutional capacity, and political structures significantly influence transparency outcomes.

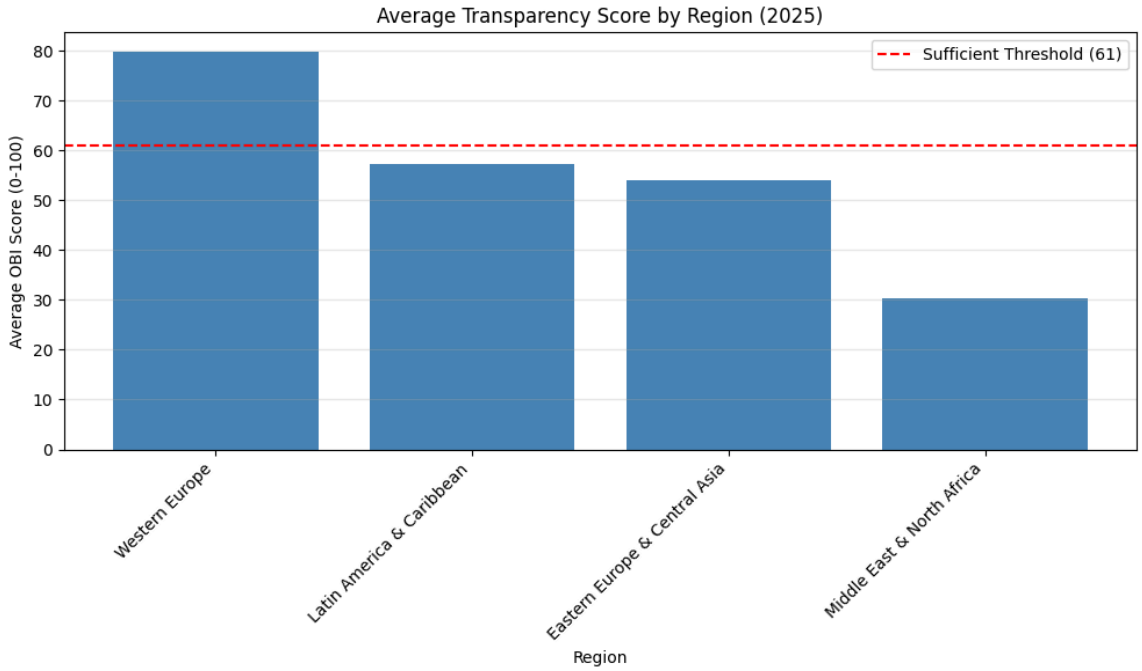

Figure 2. Average OBI by Region (2025)

Key insights:

- Western Europe consistently has the highest transparency scores
- Latin America and Eastern Europe show moderate performance
- Middle East and North Africa remain the lowest-performing region

c. Transparency and Oversight Relationship

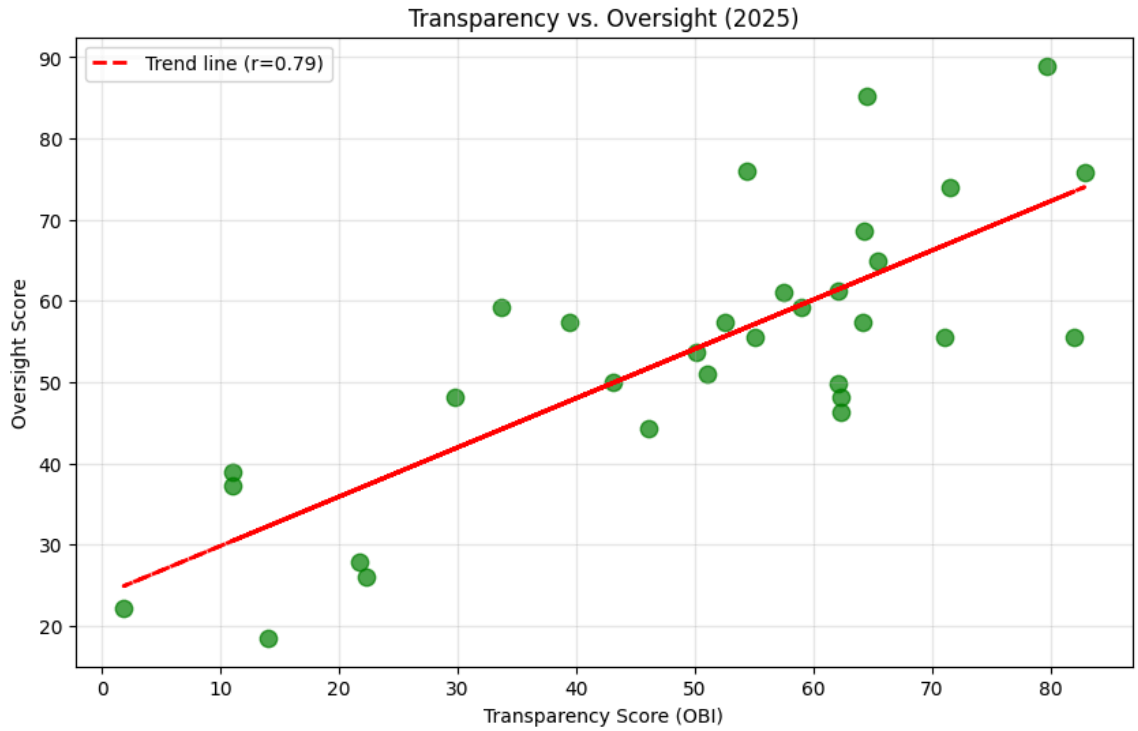

Figure 3. Transparency vs Oversight Scatter Plot

Key insights:

- A positive correlation exists between transparency and oversight
- Countries with higher transparency tend to have stronger oversight mechanisms
- This indicates that fiscal accountability components are structurally linked

d. Country-Level Changes

country	first_obi	last_obi	change
Dominican Republic	11.7	81.9	70.2
Egypt	18.5	59.0	40.5
Albania	24.9	64.2	39.3
Morocco	19.2	51.1	31.9
Nicaragua	20.6	50.1	29.5

Table 1.Top Improvers

country	first_obi	last_obi	change
North Macedonia	54.4	39.5	-14.9
Bosnia and Herzegovina	44.2	29.8	-14.5
Lebanon	32.2	21.7	-10.5
Bolivia	20.6	11.0	-9.6
Chile	72.1	64.2	-7.9

Table 2. Top Decliners

Key insights:

- Some countries show consistent improvement over time due to reforms and stronger institutions
- Others show decline, often linked to instability or weakening governance systems
- Transparency is dynamic and varies significantly across countries

e. 2021–2023 Decline Analysis

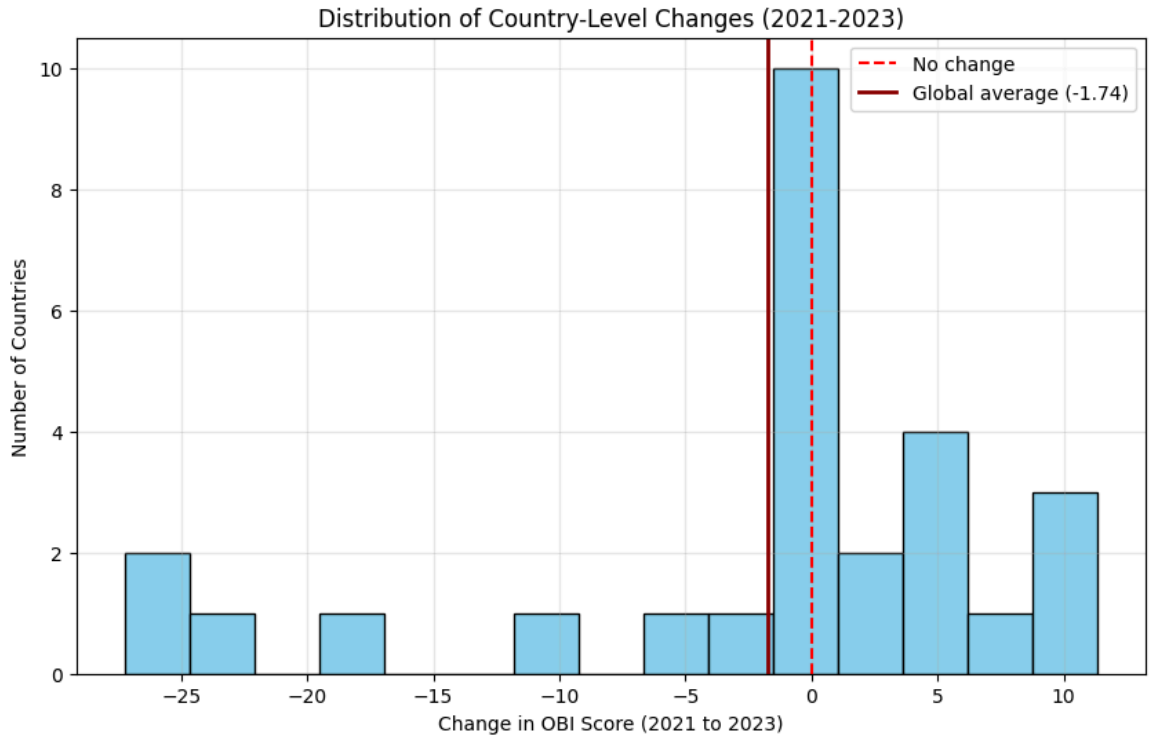

Figure 4. Distribution of Country-Level Changes (2021–2023)

country	obi_2021	obi_2023	change
Ukraine	64.8	37.6	-27.2
Tunasia	41.6	16.5	-25.1
Hungary	44.3	22.0	-22.3
El Salvador	41.3	23.5	-17.8
Bolivia	20.5	10.7	-9.7

Table 3. Countries with the Largest Drop

country	obi_2021	obi_2023	change
Algeria	3.4	14.7	11.3
Peru	61.1	71.2	10.1
Turkiye	54.6	63.9	9.3
Lebanon	8.6	17.1	8.5
Egypt	43.1	49.2	6.1

Table 4. Countries with the Largest Improvements

Key insights:

- A global decline in transparency is observed during this period
- The decline is concentrated in countries affected by political instability or democratic backsliding
- This suggests that fiscal transparency is sensitive to external political conditions

V. Conclusion

This study shows that global budget transparency has generally improved over time but remains uneven across regions and countries. Transparency is positively associated with oversight, indicating that stronger institutional systems tend to support multiple dimensions of fiscal accountability.

The analysis also highlights a significant decline between 2021 and 2023, suggesting that political instability and governance disruptions can negatively affect transparency. The partial recovery in 2025 indicates that improvements are possible when conditions stabilize.

VI. Recommendations

- Strengthen oversight institutions to support transparency
- Improve public participation in budget processes
- Encourage regional cooperation in governance reforms
- Conduct further studies using predictive modeling and clustering techniques

VII. AI Disclosure

Tool Used

- Gemini 3 Flash (Google) [March 23, 2026]
- DeepSeek (DeepSeek) [April 18, 2026]

Extent of Use

I utilized generative AI to assist in the pre-development and structural phases of this proposal. Specifically, the tool was used to:

Gemini:

- Look for possible datasets online that can be used.
- Refine the project title and research objectives based on the World Bank Open Budget Survey dataset.
- Generate a structured outline for the proposal to ensure it met all academic requirements.
- Draft initial technical descriptions for the data mining methodology (e.g., K-Means clustering and Star Schema).
- Format academic citations in accordance with APA 7th Edition standards.

Deepseek

- Assisted during the development and analysis phases of the project
- Helped troubleshoot CSV data loading issues in Jupyter Notebook
- Provided guidance on data normalization and cleaning strategies
- Explained column meanings using the DataDictionary
- Suggested analytical approaches for trend analysis, clustering, and correlation
- Assisted in interpreting output and debugging code
- Helped refine project documentation

Author's Responsibility

I have critically reviewed, edited, and verified all AI-generated content. All final decisions regarding the project's scope, technical plan, and data selection were made by me. I remain fully accountable for the accuracy and integrity of this work.

VIII. References

DeepSeek. (2026, April 18). *Assistance with data mining project analysis, troubleshooting, and methodology [Generative AI chat].* DeepSeek. https://chat.deepseek.com

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